

Name: _____

Date: _____

The coordinate grid. Every point on the grid has two numbers: its x (how far across) and its y (how far up), written (x, y) . The middle, where the lines cross, is the **origin** $(0, 0)$. Right and up are positive; left and down are negative. A *turtle* sits on the grid and follows instructions to move around and draw.

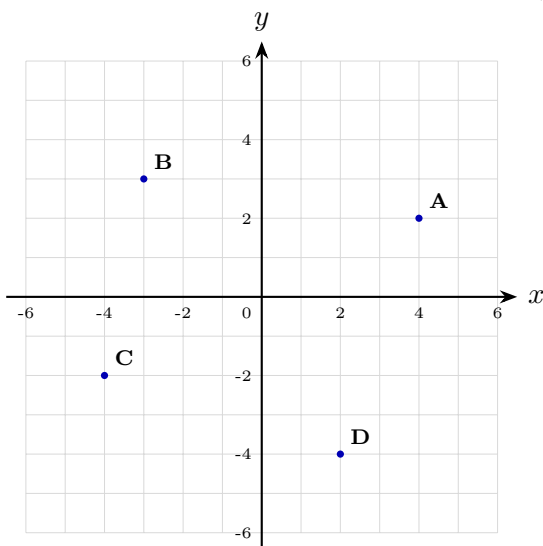
This is exactly how Scratch works. The Scratch stage is one big coordinate grid with $(0, 0)$ in the middle: x runs from -240 on the far left to 240 on the far right, and y from -180 at the bottom to 180 at the top. The blocks go to $x: y$:, **change x by**, **change y by** and **move/turn** move a sprite around that grid — the very instructions you practise below.

Command cheat sheet. Write every answer using *only* these instructions — they are the real Scratch blocks. Fill the $()$ slots with numbers.

go to $x: (x) y: (y)$ change x by (n) change y by (n) move (n) steps turn right (n) degrees turn left (n) degrees pen up pen down	jump straight to the point (x, y) move across: right is $+$, left is $-$ move up or down: up is $+$, down is $-$ go forward in the direction you are facing turn clockwise (changes facing, draws nothing) turn anticlockwise (changes facing, draws nothing) lift the pen: move without drawing put the pen down: draw as you move
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Problem 1.

Read the grid. Write the coordinates (x, y) of each labelled point.

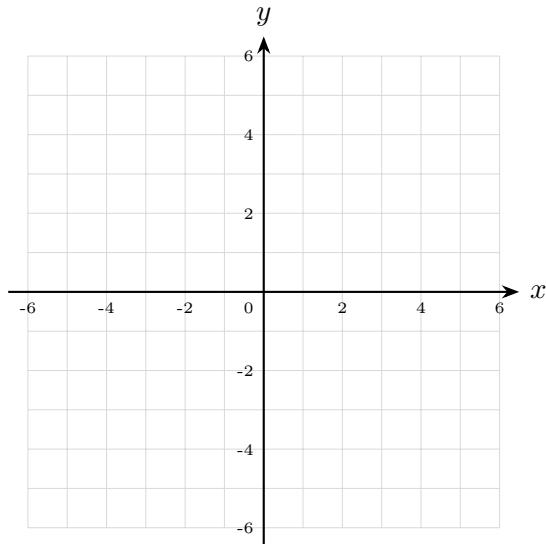


A = (____, ____) **B** = (____, ____) **C** = (____, ____) **D** = (____, ____)

Notice the signs: top-right is (+,+), top-left (-,+), bottom-left (-,-), bottom-right (+,-). Scratch points work the same way.

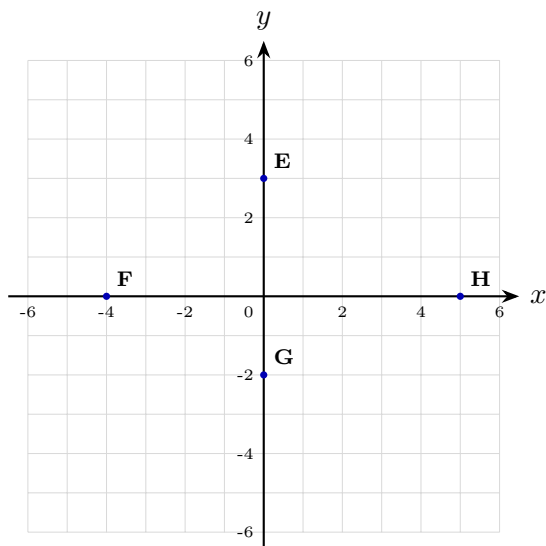
Problem 2.

Plot the points. Mark and label these points on the grid: **P**= (2,4), **Q**= (-3,2), **R**= (-2,-4), **S**= (3,-1).



Problem 3.

Read the grid. Write the coordinates (x, y) of each labelled point.



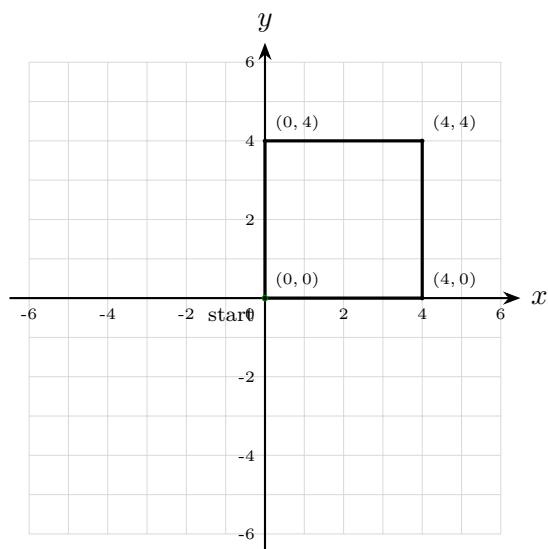
E = (____, ____) **F** = (____, ____) **G** = (____, ____) **H** = (____, ____)

These points sit on a line, not in a corner — so one of their two numbers is 0.

A point on the up-down line has $x = 0$; a point on the across line has $y = 0$. The origin $(0,0)$ is on both.

Problem 4.

Instruct the turtle. The turtle starts at the origin with its pen down. Write `go to` instructions to draw this square, visiting each corner in order.

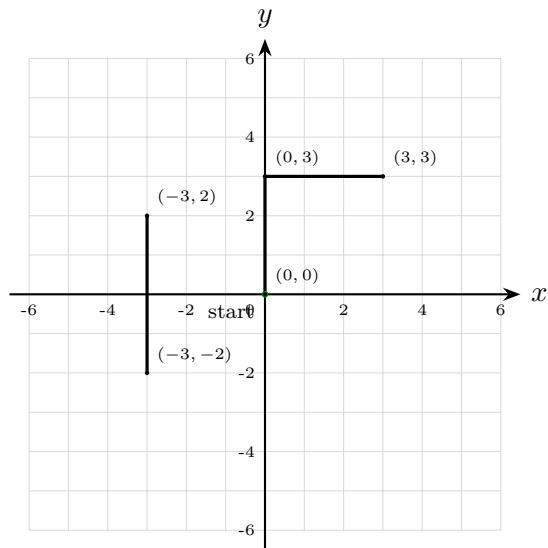


Write your instructions here:

Problem 5.

Instruct the turtle. Draw these two separate strokes. The turtle's pen must be lifted while it jumps from the end of the first stroke to the start of the second.

*Use **pen up** before the jump and **pen down** after it, or the jump will leave an unwanted line.*

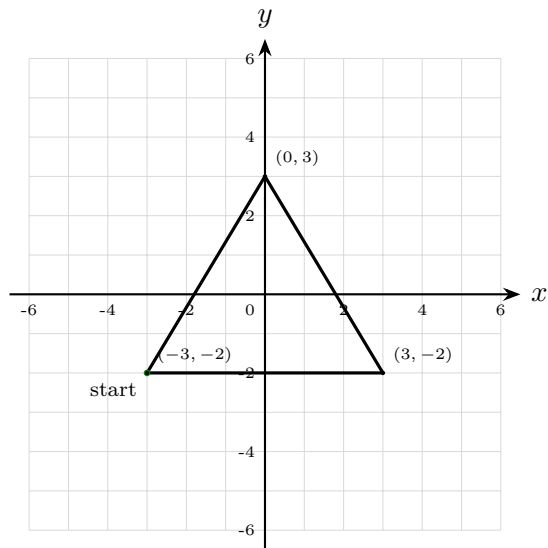


Write your instructions here:

A dashed line in the answer key means the pen was up — the turtle moved but drew nothing.

Problem 6.

Instruct the turtle. Starting at the bottom-left spot, write `go to` instructions to visit the three treasure spots and return to the start, drawing the triangle.

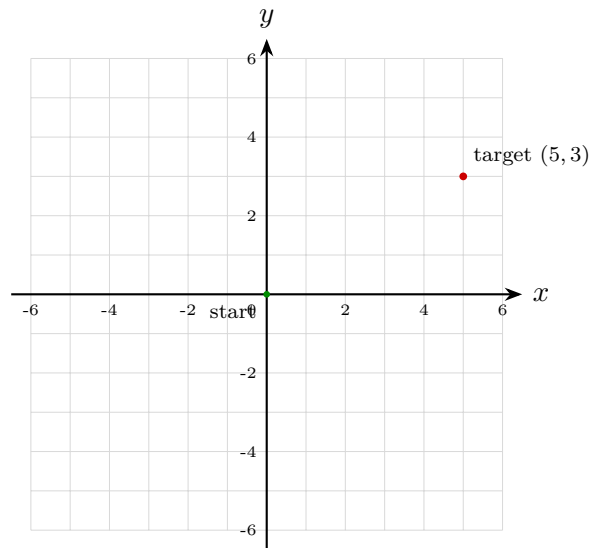


Write your instructions here:

Problem 7.

Instruct the turtle. The turtle is at the origin. Write `change x` by and `change y` by instructions to move it to the target.

change adds to where the turtle already is. How far across is the target, and how far up?

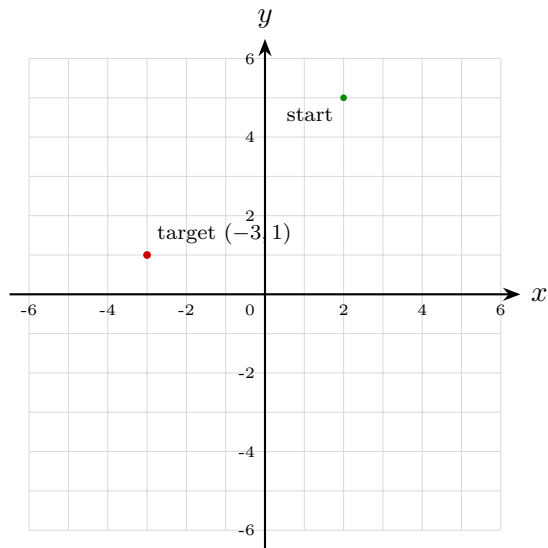


Write your instructions here:

Problem 8.

Instruct the turtle. The turtle starts at $(2, 5)$. Write **change** instructions to move it to the target. You will need negative numbers.

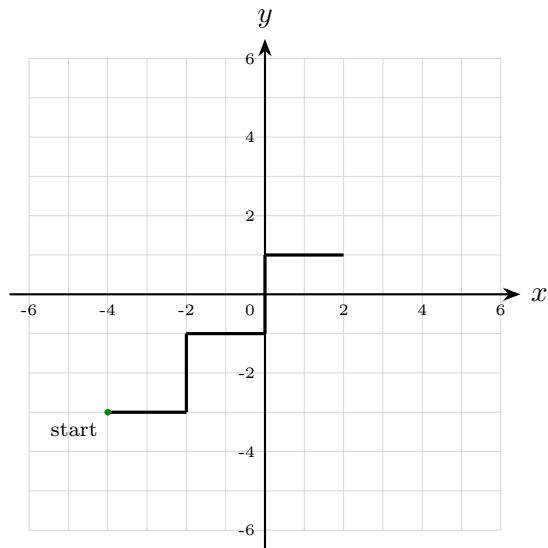
To move left, change x by a negative number; to move down, change y by a negative number.



Write your instructions here:

Problem 9.

Instruct the turtle. The turtle starts at $(-4, -3)$ with its pen down. Write `change x by` / `change y` by instructions to draw this staircase.

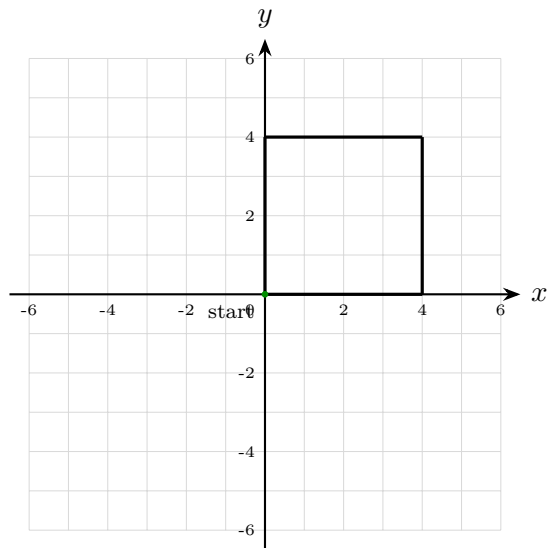


Write your instructions here:

Problem 10.

Instruct the turtle. This time use `move` and `turn` instead of `go to`. The turtle starts at the origin facing up. Write the instructions to draw this square.

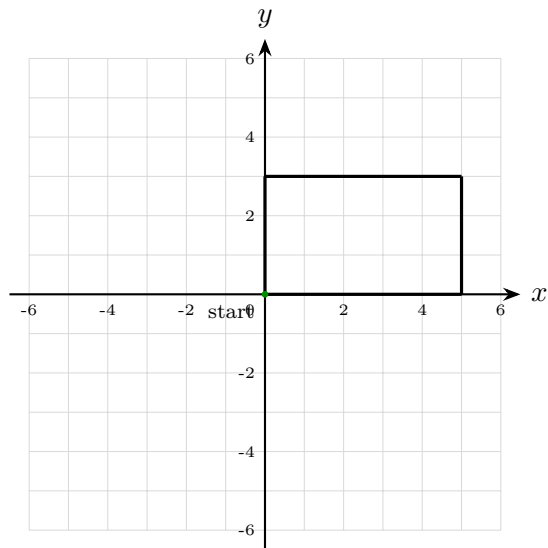
move goes forward in whatever direction the turtle is facing; turn changes that direction without drawing.



Write your instructions here:

Problem 11.

Instruct the turtle. Using move and **turn**, draw this rectangle (5 across, 3 up). The turtle starts at the origin facing up.

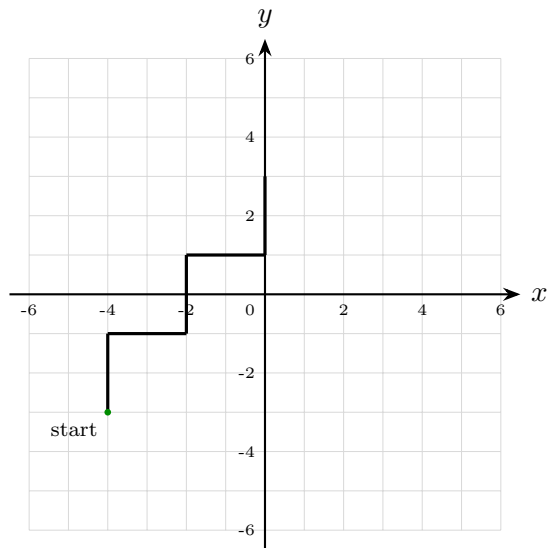


Write your instructions here:

Problem 12.

Instruct the turtle. Using `move` and `turn`, draw these steps. The turtle starts at $(-4, -3)$ facing up. Watch which way it must turn at each corner.

Turning left then right then left keeps the turtle climbing in steps rather than closing a box.

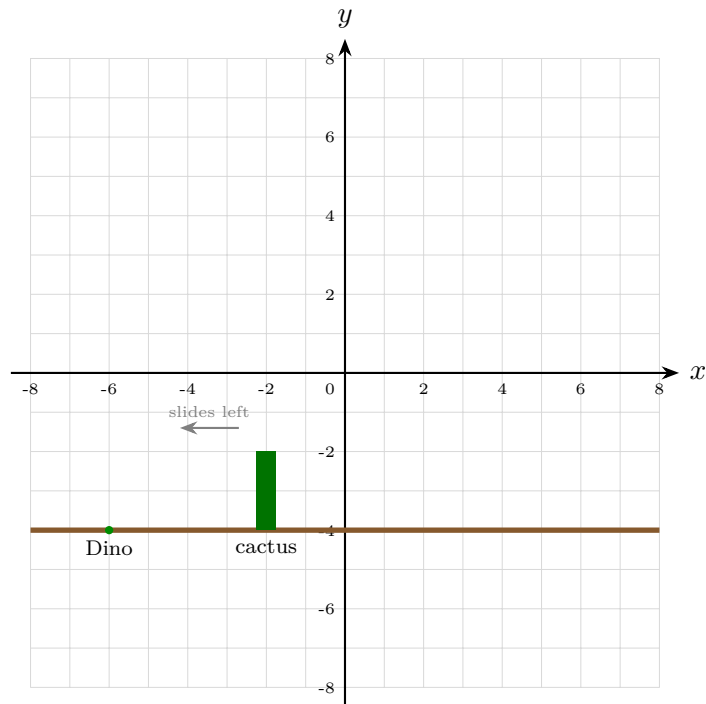


Write your instructions here:

Problem 13.

Instruct the turtle. Here is a mini Scratch stage (1 square = 20 steps). The Dino rests on the floor; a cactus slides in from the right. Write the **change y by** instructions for the Dino's jump: go up 5, then come back down 5 to land on the floor.

*The Dino never moves across — it only changes y. The cactus passes underneath using **change x by** a negative number.*



Write your instructions here:

In the real game the floor is at $y = -80$ and the cactus spawns at $x = 240$. The Dino stays at one x and only **change y by** to jump, while the cactus slides left with **change x by**. You just wrote the heart of the game.
